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Explaining the Relative Nature of General and Special Relativity

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ABSTRACT: In this paper, we show that time dilation is not a property of a single system but a comparison between two systems, defined solely by their relative spatial distribution of mass. Specifically stated, the time dilation between two systems is governed by the relative DeGerlia Inertial Density D, defined as mass divided by mean radius about the compared axis. We first exemplify this comparative nature with one of the most fundamental equivalences in General Relativity, the gravitational time-dilation equation, which computes a system's time dilation relative to flat spacetime: $d\tau/dt = \sqrt{1-r_s/r}$. We use this reference value to validate the calculation we perform using relative D alone. We then validate correct calculations for a diverse set of two-system examples, including transformed systems (similar to the gravitational time dilation above, where system 2 is a transformed version of system 1) as well as discrete system comparisons, across the classical and relativistic regimes. To conduct this demonstration, we introduce the DeGerlia Time Dilation Framework (DTDF), a simple, mathematically equivalent framework, fully defined by general relativity, that offers a second frame for visualizing the relative nature of time.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia Inertial Density, inertial density, moment of inertia, astrophysics, Schwarzschild radius, DTDF, DTD, degerlia time dilation

1 Introduction

General Relativity[1] is a model for describing our universe that is fundamentally comparative: no quantity has physical meaning without a comparator. The pace of time is no exception: the time dilation between two systems is a product of the systems' relative geometric mass distributions. Directly from the gravitational time dilation (GTD) formula, one can derive the underlying two-system logic, and ultimately, GTD is simply a comparison between a system's pace of time relative to the pace of time of a selected benchmark system, which in this case is a system of the same mass, but at its Schwarzschild radius. The formula requires only one mass and one radius, the second system being derived directly from the first. While this is the familiar representation, it obscures the comparative nature of relativity and implies that one need not compare two systems to characterize time. That implication is incorrect: there is no privileged perspective from which to characterize time. In this paper, we evaluate the physical property responsible for relativistic time dilation and explore the underlying formulation that quantifies the relative time dilation between two systems.

The relative flow of time between two systems is governed solely by their relative DeGerlia inertial density[2] D. We demonstrate this across a range of uniform spherical systems, from the classical to the relativistic regime, by calculating each system's DeGerlia time dilation from its relative D, converting each to its gravitational time dilation, and taking their ratio. In each case this matches the ratio obtained by computing each system's gravitational time dilation independently from the standard Schwarzschild form, to the precision permitted by G. Given two

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Table 1 System properties for each pair comparison case.

Property	Case 1 $m = m, \sim r_s$	Case 2 $m = m, \sim \frac{4}{3} r_s$	Case 3 $r = r$	Case 4 $\rho = \rho$	Case 5 Classical	Case 6 $I = I$	Case 7 General
m_1 (kg)	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+30	8.00000e+0	1.00000e+32	1.00000e+20
m_2 (kg)	2.00000e+30	1.00000e+30	1.00000e+24	1.00000e+27	1.00000e+0	1.00000e+30	1.00000e+30
r_1 (m)	2.97049e+3	1.98031e+3	1.00000e+8	1.00000e+8	2.00000e+0	1.00000e+7	2.00000e-7
r_2 (m)	7.00000e+8	1.00000e+8	1.00000e+8	1.00000e+7	1.00000e+0	1.00000e+8	1.00000e+7
ρ_1 (kg/m^3)	1.82161e+19	3.07406e+19	2.38732e+5	2.38732e+5	2.38732e-1	2.38732e+10	2.98416e+39
ρ_2 (kg/m^3)	1.39203e+3	2.38732e+5	2.38732e-1	2.38732e+5	2.38732e-1	2.38732e+5	2.38732e+8
I_1 (kg·m^2)	7.05907e+36	1.56865e+36	4.00000e+45	4.00000e+45	1.28000e+1	4.00000e+45	1.60000e+6
I_2 (kg·m^2)	3.92000e+47	4.00000e+45	4.00000e+39	4.00000e+40	4.00000e-1	4.00000e+45	4.00000e+43
D_1 (kg/m)	6.73289e+26	5.04972e+26	1.00000e+22	1.00000e+22	4.00000e+0	1.00000e+25	5.00000e+26
D_2 (kg/m)	2.85714e+21	1.00000e+22	1.00000e+16	1.00000e+20	1.00000e+0	1.00000e+22	1.00000e+23
r_{s1} (m)	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+3	1.18819e-26	1.48523e+5	1.48523e-7
r_{s2} (m)	2.97046e+3	1.48523e+3	1.48523e-3	1.48523e+0	1.48523e-27	1.48523e+3	1.48523e+3
$D_{\mathrm{norm},1}$	9.99991e-1	7.50001e-1	1.48523e-5	1.48523e-5	5.94093e-27	1.48523e-2	7.42617e-1
$D_{\mathrm{norm},2}$	4.24352e-6	1.48523e-5	1.48523e-11	1.48523e-7	1.48523e-27	1.48523e-5	1.48523e-4
k_s	9.99990e-1	7.50000e-1	1.48523e-5	1.48523e-5	5.94093e-27	1.48523e-2	7.42616e-1
k_d	9.99991e-1	7.50001e-1	1.48523e-5	1.48523e-5	5.94093e-27	1.48523e-2	7.42617e-1
GTD_{s1}	1-9.96838e-1	1-5.00000e-1	1-7.42619e-6	1-7.42619e-6	1-2.97046e-27	1-7.45394e-3	1-4.92670e-1
GTD_{s2}	1-2.12176e-6	1-7.42619e-6	1-7.42616e-12	1-7.42616e-8	1-7.42616e-28	1-7.42619e-6	1-7.42644e-5
GTD_{d1}	1-9.96948e-1	1-5.00001e-1	1-7.42619e-6	1-7.42619e-6	1-2.97047e-27	1-7.45395e-3	1-4.92670e-1
GTD_{d2}	1-2.12176e-6	1-7.42619e-6	1-7.42617e-12	1-7.42617e-8	1-7.42617e-28	1-7.42619e-6	1-7.42644e-5
DTD_1	9.99995e-1	8.66026e-1	3.85387e-3	3.85387e-3	7.70774e-14	1.21870e-1	8.61752e-1
DTD_2	2.05998e-3	3.85387e-3	3.85387e-6	3.85387e-4	3.85387e-14	3.85387e-3	1.21870e-2

Table 2 Derived ratios across cases.

Ratio	Case 1 $m = m, \sim r_s$	Case 2 $m = m, \sim \frac{4}{3} r_s$	Case 3 $r = r$	Case 4 $\rho = \rho$	Case 5 Classical	Case 6 $I = I$	Case 7 General
$\mathrm{DTD}_1/\mathrm{DTD}_2$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$\sqrt{D_1/D_2}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$\sqrt{(m_1 r_2)/(m_2 r_1)}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$\sqrt{(I_1/I_2)(r_2/r_1)^3}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$(m_1 r_2)/(m_2 r_1)$	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	4.00000e+0	1.00000e+3	5.00000e+3
D_1/D_2	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	4.00000e+0	1.00000e+3	5.00000e+3
$(I_1/I_2)(r_2/r_1)^3$	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	4.00000e+0	1.00000e+3	5.00000e+3
$\mathrm{GTD}_{s1}/\mathrm{GTD}_{s2}$	1-9.96838e-1	1-4.99996e-1	1-7.42618e-6	1-7.35193e-6	1-2.22785e-27	1-7.44657e-3	1-4.92632e-1
$\mathrm{GTD}_{d1}/\mathrm{GTD}_{d2}$	1-9.96948e-1	1-4.99997e-1	1-7.42619e-6	1-7.35193e-6	1-2.22785e-27	1-7.44658e-3	1-4.92633e-1
$\sqrt{1-\mathrm{DTD}_1^2}/\sqrt{1-\mathrm{DTD}_2^2}$	1-9.96948e-1	1-4.99997e-1	1-7.42619e-6	1-7.35193e-6	1-2.22785e-27	1-7.44658e-3	1-4.92633e-1
I_1/I_2	1.80078e-11	3.92163e-10	1.00000e+6	1.00000e+5	3.20000e+1	1.00000e+0	4.00000e-38
$(I_1/I_2)^{1/2}$	4.24356e-6	1.98031e-5	1.00000e+3	3.16228e+2	5.65685e+0	1.00000e+0	2.00000e-19
$(\rho_1/\rho_2)^{1/5}(r_1/r_2)$	7.09728e-3	1.31430e-2	1.58489e+1	1.00000e+1	2.00000e+0	1.00000e+0	3.31445e-8
$(I_1/I_2)^{1/5}$	7.09728e-3	1.31430e-2	1.58489e+1	1.00000e+1	2.00000e+0	1.00000e+0	3.31445e-8
m_1/m_2	1.00000e+0	1.00000e+0	1.00000e+6	1.00000e+3	8.00000e+0	1.00000e+2	1.00000e-10
$\sqrt{m_1/m_2}$	1.00000e+0	1.00000e+0	1.00000e+3	3.16228e+1	2.82843e+0	1.00000e+1	1.00000e-5
r_1/r_2	4.24356e-6	1.98031e-5	1.00000e+0	1.00000e+1	2.00000e+0	1.00000e-1	2.00000e-14
$\sqrt{r_1/r_2}$	2.05999e-3	4.45007e-3	1.00000e+0	3.16228e+0	1.41421e+0	3.16228e-1	1.41421e-7

systems' masses and radii, their relative time dilation is determined by their relative spatial distribution of mass alone, a relation already contained in general relativity.

2 Experimental Model

The experimental model employed in this study comprises several components. The framework is derived directly from general relativity and is mathematically equivalent to established general-relativistic calculations. It neither relies upon nor introduces any principle not already established within the theories of general and special relativity; this paper leverages a framing of the existing theory of relativity for illustrative purposes.

We begin by introducing the DeGerlia Framework and then demonstrate its complete mathematical equivalence with general relativity. We then work through a series of examples, each demonstrating a different comparative case, and in each case, we validate the result using standard gravitational time-dilation calculations.

These examples demonstrate the two-system nature of all relativistic calculations. Relativity is fundamentally concerned with how quantities appear relative to one another; gravitational time dilation, for instance, reflects the comparison of a system with an identical system at its Schwarzschild radius, the collapse-condition radius r_s.

3 The DeGerlia Time Dilation Framework

3.1 Definitions, Formulas and Constants Used Herein

Symbol	Definition / Value
c	Light Speed $299\,792\,458\,\mathrm{m\,s^{-1}}$
D	DeGerlia Inertial Density m/r, where r is the mean radius about a selected axis
D_{crit}	DeGerlia Threshold $c^2/2G = 6.73295\mathrm{e}26\,\mathrm{kg/m}$
DTD	DeGerlia Time Dilation $\sqrt{k}$
G	Gravitational Constant $6.67430\mathrm{e}{-11} \pm 1.5\mathrm{e}{-15}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$
GTD	Gravitational Time Dilation $d\tau/dt = \sqrt{1-k}$
k	Linear Scale Factor $v^2/c^2 = r_s/r = D/D_{\mathrm{crit}}$
k_D	Linear Scale Factor (DeGerlia inertial density) D/D_{crit}
k_s	Linear Scale Factor (Schwarzschild) r_s/r
k_v	Linear Scale Factor (velocity) v^2/c^2
P	Inertial Density $I/V = D \cdot 3/(10\pi)$
r_s	Schwarzschild Radius $2Gm/c^2$

3.2 What is the DeGerlia Time Dilation Framework?

First and foremost, the entirety of the principles described herein represents no deviation from general relativity. It represents a mathematically equivalent reframing useful in certain relativistic calculations. The author has simply asserted a complement to GTD, where the form of time dilation is equal to $\sqrt{k}$, instead of $\sqrt{1-k}$.

While equivalent to GTD mathematically, the DTD form offers a different perspective on the same physical phenomenon. It is a complementary perspective that aids in visualizing the underlying physics:

- One can convert between GTD and DTD at any time. GTD is always recoverable.
- DTD is mathematically very easy to use and visualize, requiring only the ratio of mass to radius in comparison to a static constant to perform gravitational time dilation calculations.
- DTD allows for direct calculation of time dilation between two systems, without the need to reference a third system (the Schwarzschild condition).
- DTD returns finite output over the entire range of finite input.
- DTD represents flat spacetime as an ideal state that can only be approached.
- DTD represents the Schwarzschild condition as the point where the gravitational escape velocity exceeds the speed of light, and an event horizon forms. While we have observed black holes, the asymptotically approached ideal state of zero time dilation is not attainable.
- DTD reflects the singularity state as an asymptotic approach to $r = 0$.

3.3 Inertia and Moment of Inertia

In classical physics, inertia dictates how a system's motion changes in response to an applied force. This applies to both linear and rotational motion. In rotational dynamics, the "moment of inertia"[3] I dictates how the angular motion of a system about a specific axis changes when a torque is applied, and is defined as the sum of each mass element multiplied by the square of its distance from the axis of rotation.

$$\text{Moment of Inertia: } I = \sum_i m_i r_i^2 \tag{1}$$

Using Equation (1), one can determine how a system's motion changes when energy is applied about an axis (torque). As it pertains to linear motion, inertia (I) is characterized solely by the system's mass.

Inertia and moment of inertia have several notable properties.

1. Inertia is dimensional: it exists in, and has a component within, each of the three spatial dimensions.
2. Inertia is multimodal: it may be linear or rotational. Linear inertia reflects the mass alone, whereas rotational inertia reflects the mass times the square of its radius about an axis.
3. The moment of inertia is additive about a given axis: if a system is divided into parts and each part's moment of inertia is measured about that same axis, the total is unchanged.
4. Inertia relates force to motion: it determines the change in motion produced by an applied force or torque, making it the bridge between the two. Lower inertia yields greater motion for a given force.

3.4 General Relativity and Time Dilation

General Relativity provides the formula for calculating a system's time dilation relative to "flat space-time".

$$d\tau/dt = \sqrt{1-k} \tag{2}$$

where $k = r_s/r = 2Gm/(rc^2)$, m is the system's mass, and r its radius.

The equation requires only a single system's mass and radius, yet k is itself a ratio of two radii: the system's radius r and the Schwarzschild radius r_s (11) of that same mass. The calculation is therefore a comparison of two systems: the system as given, and the same mass at its Schwarzschild radius.

3.5 Definition of a System

A "system" is defined as any discrete collection of particles and the space between them, or equivalently, any unit of space and the particles contained within it. The framework regards the universe not as a clean separation of discrete material systems separated by the vacuum of space, but as a gradient of material density pervasive throughout. The most extreme densities reside not within near-scale systems but within the cores of highly dense gravitational objects far from the human scale, such as black holes or subatomic particles, and form a gradient extending to the near-scale "vacuum of space" at which another gravitational source begins to dominate. The framework imposes no requirement for uniformity or symmetry, a clean boundary separating matter from vacuum, or any particular density profile. It requires only a mass and a radius about a chosen axis. A system may be specified, for example, as the Earth out to the mean radius of its surface. For a body such as Jupiter, which lacks a clear boundary separating surface from atmosphere, the system may instead be specified out to a chosen radius or density. For a large system, a spherical sample may be taken from within it to characterize the behavior of particular regions. There can be no system modeled under General Relativity that lacks either mass or radius.

3.6 System Time

For a whole system (as distinct from any subsystem), and from that whole system's perspective, the pace of time is static and all other systems are perceived relative to that benchmark. A subsystem possesses distinct physical properties and consequently exhibits behavior different from that of the system as a whole. In general relativity, time dilation scales precisely with the relative mass-to-radius ratio; a part of a system, therefore, cannot be expected to share the pace of time of the whole.

3.7 Inertial Density P

Inertial density P as shown in (4) is a fundamental system property[4] characterizing the distribution of inertia across any system, whether in the linear or rotational context, in any direction, or about any axis. If we were to redefine classical density as I/V (inertia/volume), because inertia is mass for one-dimensional motion, this would remain consistent with the classical definition of density, where $\rho = m/V$.

Inertial density characterizes how a system's mass is distributed relative to its axis, for a given mass and volume. Consider the configuration with the greatest moment of inertia for a given volume: a thin shell, in which all mass resides at the radius, displaced as far from the axis as the geometry permits. A soap bubble exemplifies this configuration; per unit volume it possesses substantial inertia despite its low mass, representing the maximum-inertia structure for its volume. Conversely, a configuration of equal radius with its mass concentrated as an approximate point mass at the center represents the opposite limit: as the mass concentrates toward the center, the moment of inertia approaches zero. The uniform solid sphere lies between these limits. What determines the inertia about a given axis is the mean radius of the mass distribution about that axis; the internal density gradient that produces a given mean radius does not enter, just as ordinary density averages over a system's internal structure and the density of any subset is obtained by evaluating that subset directly.

3.8 Mathematics of Inertial Density

To derive the inertial density relationship, we begin with the established relationship between mass and density ρ as shown in (3):

$$\rho = \frac{m}{V} \tag{3}$$

where ρ is density, m is mass, and V is volume. Because mass is equivalent to linear inertia, and because density is m/V, we define inertial density P in terms of moment of inertia over volume:

$$\text{Inertial Density}: P = \frac{I}{V} \tag{4}$$

where inertial density P is equal to inertia I over volume V. Note, the moment of inertia I for a system may be distinct about every axis for systems that are not uniform[5]. For a uniform sphere[6], we substitute $I = \frac{2}{5}mr^2$ and $V = \frac{4}{3}\pi r^3$:

$$P = \frac{I}{V} = \frac{\frac{2}{5}mr^2}{\frac{4}{3}\pi r^3} = \frac{3m}{10\pi r} \tag{5}$$

Which can be simplified as follows:

$$P = \frac{m}{r} \times \frac{3}{10\pi} \tag{6}$$

$$P = \frac{m}{r} \times 9.549\,296\,585\text{e}{-2} \tag{7}$$

Therefore, for spherical systems, P can be calculated as the ratio of mass to radius, multiplied by the conversion factor $\frac{3}{10\pi}$ as shown in (7).

3.9 DeGerlia Inertial Density D

For all spherical systems, m/r multiplied by the geometric factor $3/(10\pi)$ yields the inertial density. ***DeGerlia inertial density*** is a simplified analog of inertial density I/V, equal to the mass-to-radius ratio m/r with the $3/(10\pi)$ geometric conversion factor omitted:

$$\boxed{\text{DeGerlia Inertial Density: } D = \frac{m}{r}} \tag{8}$$

DeGerlia inertial density (8) $D = m/r$ is related to inertial density P by the geometric factor $3/(10\pi)$ as shown in (9):

$$P = \frac{3}{10\pi} \times D \tag{9}$$

In a linear context, inertia is mass, and inertial density reduces directly to mass density $\rho = m/V$. From this perspective, inertial density is the rotational analog of mass density, paralleling the relationship between $F = ma$ and $\tau = I\alpha$. Equivalently, mass density is the linear interpretation of inertial density.

Just as ordinary density ($\rho = m/V$) is linear inertia (mass) distributed over volume, DeGerlia inertial density is rotational inertia distributed over volume. As stated in the prior work, *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

Because D is almost always used in a ratio that cancels all geometric constants, this metric is typically preferred over the geometrically accurate inertial density P for ease of use.

3.10 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia inertial density D, rather than as a value that must be recalculated for every mass.

Setting $r = r_s$ (the Schwarzschild condition), the DeGerlia inertial density D becomes the threshold D_{crit}:

$$D_{\text{crit}} = \frac{m}{r_s} \tag{10}$$

Recall the Schwarzschild radius[7]:

$$r_s = \frac{2Gm}{c^2} \tag{11}$$

Substitute r_s into the expression for D_{crit}:

$$D_{\text{crit}} = \frac{m}{\frac{2Gm}{c^2}} = \frac{mc^2}{2Gm} = \frac{c^2}{2G} \tag{12}$$

Insert the fundamental constants[8], $c = 2.99792458\text{e}8\,\text{m}\,\text{s}^{-1}$ and $G = 6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$:

$$\begin{aligned} D_{\text{crit}} &= \frac{(2.99792458\text{e}8)^2}{2 \times 6.67430\text{e}{-11}}\,\text{kg/m} \\ &= \frac{8.98755179\text{e}16}{1.334860\text{e}{-10}} \\ &= 6.73295\text{e}26\,\text{kg/m} \end{aligned}$$

$$\boxed{\text{DeGerlia Threshold: } D_{\text{crit}} = 6.73295\text{e}26\,\text{kg/m}} \tag{13}$$

The DeGerlia Threshold (13) is the universal constant value of DeGerlia inertial density D at which the escape velocity equals the speed of light: the mass-over-radius threshold at which gravitational collapse occurs. Note this value is limited by the precision of the gravitational constant[9], $G = 6.67430\text{e}{-11} \pm 1.5\text{e}{-15}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$. Just as the Schwarzschild radius, the DeGerlia threshold is universally applicable to spherical gravitational systems; examples include black holes or neutron stars, a spherical region of an interstellar cloud, or even a galactic cluster.

Rapid evaluation of the black hole condition. For a single system, divide its mass by its radius and compare the result with the DeGerlia Threshold (13), $D_{\text{crit}} = 6.73295\text{e}26\,\text{kg/m}$. If their ratio equals 1, the system is at the black hole condition; if it is less than 1, no collapse has occurred.

3.11 DeGerlia Time Dilation

DeGerlia Time Dilation (DTD) (14) is the Pythagorean complement (15) of gravitational time dilation:

$$\text{GTD} = \sqrt{1-k}$$

$$\text{DTD} = \sqrt{k} \tag{14}$$

$$\text{GTD}^2 + \text{DTD}^2 = 1 \tag{15}$$

where k is the linear scale factor.

3.12 The Linear Scale Factor k

The linear scale factor k is a common element among all established forms of time dilation. Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k}$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the linear scale factor between the two systems. Specific forms of k follow:

- $k = r_s/r$, the standard Schwarzschild condition.
- $k = D/D_{\text{crit}} = (m/r)/(6.73295\text{e}26\,\text{kg/m})$, via the DeGerlia inertial density against its threshold.
- $k = v^2/c^2$, as used in special relativity.

4 Relative Time Dilation Examples

Tables 1 and 2 present the complete system properties and derived ratios for seven system pairs spanning the cases that follow: equal mass at the Schwarzschild radius, equal mass at four-thirds the Schwarzschild radius, equal radius, equal density, a classical case, equal moment of inertia, and a general unconstrained case.

Across every case, the time-dilation ratio computed via the DeGerlia inertial density ratio agrees with the gravitational time-dilation ratio computed independently from the standard Schwarzschild form, to the limit of precision imposed by G. No case requires a separate formula. The worked examples in the following subsections show the full calculations, case by case.

4.1 Case 1: ($m_1 = m_2$, $r_1 \approx r_s$)

The two identical masses cancel, reducing the DeGerlia inertial density ratio to a ratio of radii. Note that this reflects the k calculation for gravitational time dilation, where we compare with r_s. Because $GTD = 0$ exactly at r_s, we set the comparison radius to $1.00001\,r_s$, an "arbitrarily" small amount above r_s.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	2.000 00e30	2.000 00e30
r (m)	2.970 49e3	7.000 00e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$
$$k_1 = \frac{2.00000\text{e}30\,\text{kg}}{(2.97049\text{e}3\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 9.9999068441\text{e}{-1}$$
$$DTD_1 = \sqrt{9.9999068441\text{e}{-1}} = 9.9999534220\text{e}{-1}$$
$$k_2 = \frac{2.00000\text{e}30\,\text{kg}}{(7.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 4.24352\text{e}{-6}$$
$$DTD_2 = \sqrt{4.24352\text{e}{-6}} = 2.05998\text{e}{-3}$$
$$\frac{DTD_1}{DTD_2} = \frac{9.9999534220\text{e}{-1}}{2.05998\text{e}{-3}} = 4.85439\text{e}2$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(9.9999534220\text{e}{-1})^2}}{\sqrt{1-(2.05998\text{e}{-3})^2}}$$
$$= \frac{\sqrt{1-9.9999068441\text{e}{-1}}}{\sqrt{1-4.24352\text{e}{-6}}}$$
$$= \frac{\sqrt{9.31559\text{e}{-6}}}{\sqrt{9.9999575648\text{e}{-1}}}$$
$$= \frac{3.05214\text{e}{-3}}{9.9999787824\text{e}{-1}} = 3.05215\text{e}{-3}$$
$$= 1 - 9.9694785\text{e}{-1}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2Gm}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(2.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 2.97046\text{e}3\,\text{m}$$

Calculate k values from the r_s/r ratio:

$$k_1 = \frac{r_{s1}}{r_1} = \frac{2.97046\text{e}3\,\text{m}}{2.97049\text{e}3\,\text{m}} = 9.9999000010\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{r_2} = \frac{2.97046\text{e}3\,\text{m}}{7.00000\text{e}8\,\text{m}} = 4.24352\text{e}{-6}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{9.99990\text{e}{-6}} = 3.16226\text{e}{-3} = 1 - 9.9683774\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.9999575648\text{e}{-1}} = 9.9999787824\text{e}{-1}$$
$$= 1 - 2.12176\text{e}{-6}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{3.16226\text{e}{-3}}{9.9999787824\text{e}{-1}} = 3.16227\text{e}{-3}$$
$$= 1 - 9.9683773\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 9.9683773\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 9.9694785\text{e}{-1} \quad (\text{via } D_{\text{crit}})$$

4.2 Case 2: ($m_1 = m_2$, $r_1 = \frac{4}{3} r_s$)

The two identical masses cancel, reducing the DeGerlia inertial density ratio to a ratio of radii. Note: This reflects the k calculation for gravitational time dilation. However, in this case, instead of comparing with r_s, we compare with $\frac{4}{3} r_s$, which corresponds to a gravitational time dilation of $GTD = 5.00000\text{e}{-1}$.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.000 00e30	1.000 00e30
r (m)	1.980 31e3	1.000 00e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$
$$k_1 = \frac{1.00000\text{e}30\,\text{kg}}{(1.98031\text{e}3\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 7.50001\text{e}{-1}$$
$$DTD_1 = \sqrt{7.50001\text{e}{-1}} = 8.66026\text{e}{-1}$$
$$k_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$
$$\frac{DTD_1}{DTD_2} = \frac{8.66026\text{e}{-1}}{3.85387\text{e}{-3}} = 2.24716\text{e}2$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(8.66026\text{e}{-1})^2}}{\sqrt{1-(3.85387\text{e}{-3})^2}}$$
$$= \frac{\sqrt{1-7.50001\text{e}{-1}}}{\sqrt{1-1.48523\text{e}{-5}}}$$
$$= \frac{\sqrt{2.49999\text{e}{-1}}}{\sqrt{9.999851477\text{e}{-1}}}$$
$$= \frac{4.99999\text{e}{-1}}{9.9999257381\text{e}{-1}} = 5.00003\text{e}{-1}$$
$$= 1 - 4.99997\text{e}{-1}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2Gm}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$

Calculate k values from the r_s/r ratio:

$$k_1 = \frac{r_{s1}}{r_1} = \frac{1.48523\text{e}3\,\text{m}}{1.98031\text{e}3\,\text{m}} = 7.50000\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{r_2} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{2.50000\text{e}{-1}} = 5.00000\text{e}{-1} = 1 - 5.00000\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1}$$
$$= 1 - 7.42619\text{e}{-6}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{5.00000\text{e}{-1}}{9.9999257381\text{e}{-1}} = 5.00004\text{e}{-1}$$
$$= 1 - 4.99996\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 4.99996\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 4.99997\text{e}{-1} \quad (\text{via } D_{\text{crit}})$$

4.3 Case 3: ($r_1 = r_2$)

When the two systems share the same radius, the radius terms cancel. The Lorentz factor [10] has the same $\sqrt{1-k}$ form, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit canceling.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.000 00e30	1.000 00e24
r (m)	1.000 00e8	1.000 00e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$
$$k_1 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$
$$DTD_1 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$
$$k_2 = \frac{1.00000\text{e}24\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-11}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-11}} = 3.85387\text{e}{-6}$$
$$\frac{DTD_1}{DTD_2} = \frac{3.85387\text{e}{-3}}{3.85387\text{e}{-6}} = 1.00000\text{e}3$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(3.85387\text{e}{-3})^2}}{\sqrt{1-(3.85387\text{e}{-6})^2}}$$
$$= \frac{\sqrt{1-1.48523\text{e}{-5}}}{\sqrt{1-1.48523\text{e}{-11}}}$$
$$= \frac{\sqrt{9.999851477\text{e}{-1}}}{\sqrt{9.999999999851477\text{e}{-1}}}$$
$$= \frac{9.9999257381\text{e}{-1}}{9.9999999999257383\text{e}{-1}} = 9.9999257381\text{e}{-1}$$
$$= 1 - 7.42619\text{e}{-6}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2Gm_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$
$$r_{s2} = \frac{2Gm_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}24\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}{-3}\,\text{m}$$

Calculate k values from the r_s/r ratio:

$$k_1 = \frac{r_{s1}}{r_1} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$
$$k_2 = \frac{r_{s2}}{r_2} = \frac{1.48523\text{e}{-3}\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-11}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1}$$
$$= 1 - 7.42619\text{e}{-6}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.999999999851477\text{e}{-1}}$$
$$= 9.9999999999257384\text{e}{-1} = 1 - 7.42616\text{e}{-12}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{9.9999257381\text{e}{-1}}{9.9999999999257384\text{e}{-1}}$$
$$= 9.9999257382\text{e}{-1} = 1 - 7.42618\text{e}{-6}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 7.42618\text{e}{-6} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 7.42619\text{e}{-6} \quad (\text{via } D_{\text{crit}})$$

4.4 Case 4: ($\rho_1 = \rho_2$)

When the two systems share the same density, $m \propto r^3$, so $m_1/m_2 = (r_1/r_2)^3$.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.000 00e30	1.000 00e27
r (m)	1.000 00e8	1.000 00e7

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$
$$k_1 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$
$$DTD_1 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$
$$k_2 = \frac{1.00000\text{e}27\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-7}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-7}} = 3.85387\text{e}{-4}$$
$$\frac{DTD_1}{DTD_2} = \frac{3.85387\text{e}{-3}}{3.85387\text{e}{-4}} = 1.00000\text{e}1$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(3.85387\text{e}{-3})^2}}{\sqrt{1-(3.85387\text{e}{-4})^2}}$$
$$= \frac{\sqrt{1-1.48523\text{e}{-5}}}{\sqrt{1-1.48523\text{e}{-7}}}$$
$$= \frac{\sqrt{9.999851477\text{e}{-1}}}{\sqrt{9.99999851477\text{e}{-1}}}$$
$$= \frac{9.9999257381\text{e}{-1}}{9.999999257383\text{e}{-1}} = 9.9999264807\text{e}{-1}$$
$$= 1 - 7.35193\text{e}{-6}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2Gm_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$
$$r_{s2} = \frac{2Gm_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}27\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\,\text{m}$$

Calculate k values from the r_s/r ratio:

$$k_1 = \frac{r_{s1}}{r_1} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$
$$k_2 = \frac{r_{s2}}{r_2} = \frac{1.48523\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-7}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1}$$
$$= 1 - 7.42619\text{e}{-6}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.99999851477\text{e}{-1}} = 9.999999257384\text{e}{-1}$$
$$= 1 - 7.42616\text{e}{-8}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{9.9999257381\text{e}{-1}}{9.999999257384\text{e}{-1}}$$
$$= 9.9999264807\text{e}{-1} = 1 - 7.35193\text{e}{-6}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 7.35193\text{e}{-6} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 7.35193\text{e}{-6} \quad (\text{via } D_{\text{crit}})$$

4.5 Case 5: (Classical)

This is a static-density, isometric case that follows the same model as Case 4.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	8.00000	1.00000
r (m)	2.00000	1.00000

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$

$$k_1 = \frac{8.00000\,\text{kg}}{(2.00000\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 5.94093\text{e}{-27}$$

$$DTD_1 = \sqrt{5.94093\text{e}{-27}} = 7.70774\text{e}{-14}$$

$$k_2 = \frac{1.00000\,\text{kg}}{(1.00000\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-27}$$

$$DTD_2 = \sqrt{1.48523\text{e}{-27}} = 3.85387\text{e}{-14}$$

$$\frac{DTD_1}{DTD_2} = \frac{7.70774\text{e}{-14}}{3.85387\text{e}{-14}} = 2.00000$$

Calculate the GTD ratio from DTD:

$$\begin{aligned}
\frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(7.70774\text{e}{-14})^2}}{\sqrt{1-(3.85387\text{e}{-14})^2}} \\
&= \frac{\sqrt{1-5.94093\text{e}{-27}}}{\sqrt{1-1.48523\text{e}{-27}}} \\
&= \frac{\sqrt{9.999\,999\,999\,999\,999\,999\,999\,999\,940\,590\,7\text{e}{-1}}}{\sqrt{9.999\,999\,999\,999\,999\,999\,999\,999\,985\,147\,7\text{e}{-1}}} \\
&= \frac{9.999\,999\,999\,999\,999\,999\,999\,999\,970\,295\,3\text{e}{-1}}{9.999\,999\,999\,999\,999\,999\,999\,999\,992\,573\,83\text{e}{-1}} \\
&= 9.999\,999\,999\,999\,999\,999\,999\,999\,977\,721\,5\text{e}{-1} \\
&= 1 - 2.22785\text{e}{-27}
\end{aligned}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$\begin{aligned}
r_{s1} &= \frac{2Gm_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(8.00000\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2} \\
&= 1.18819\text{e}{-26}\,\text{m} \\
r_{s2} &= \frac{2Gm_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2} \\
&= 1.48523\text{e}{-27}\,\text{m}
\end{aligned}$$

Calculate k values from the r_s/r ratio:

$$k_1 = \frac{r_{s1}}{r_1} \qquad = \frac{1.18819\text{e}{-26}\,\text{m}}{2.00000\,\text{m}} \qquad = 5.94093\text{e}{-27}$$

$$k_2 = \frac{r_{s2}}{r_2} \qquad = \frac{1.48523\text{e}{-27}\,\text{m}}{1.00000\,\text{m}} \qquad = 1.48523\text{e}{-27}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
GTD_{s1} &= \sqrt{1-k_1} \\
&= \sqrt{9.999\,999\,999\,999\,999\,999\,999\,999\,940\,590\,7\text{e}{-1}} \\
&= 9.999\,999\,999\,999\,999\,999\,999\,999\,970\,295\,4\text{e}{-1} \\
&= 1 - 2.97046\text{e}{-27} \\
GTD_{s2} &= \sqrt{1-k_2} \\
&= \sqrt{9.999\,999\,999\,999\,999\,999\,999\,999\,985\,147\,7\text{e}{-1}} \\
&= 9.999\,999\,999\,999\,999\,999\,999\,999\,992\,573\,84\text{e}{-1} \\
&= 1 - 7.42616\text{e}{-28} \\
\frac{GTD_{s1}}{GTD_{s2}} &= \frac{9.999\,999\,999\,999\,999\,999\,999\,999\,970\,295\,4\text{e}{-1}}{9.999\,999\,999\,999\,999\,999\,999\,999\,992\,573\,84\text{e}{-1}} \\
&= 9.999\,999\,999\,999\,999\,999\,999\,999\,977\,721\,5\text{e}{-1} \\
&= 1 - 2.22785\text{e}{-27}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 2.22785\text{e}{-27} \quad (\text{via } r_s)$$

$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 2.22785\text{e}{-27} \quad (\text{via } D_{\text{crit}})$$

4.6 Case 6: ($I_1 = I_2$)

When the two systems share the same moment of inertia, $m_1 r_1^2 = m_2 r_2^2$, so $m_1/m_2 = (r_2/r_1)^2$. The DeGerlia inertial density ratio reduces to the cube of the radius ratio.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.00000e32	1.00000e30
r (m)	1.00000e7	1.00000e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$

$$k_1 = \frac{1.00000\text{e}32\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-2}$$

$$DTD_1 = \sqrt{1.48523\text{e}{-2}} = 1.21870\text{e}{-1}$$

$$k_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$

$$DTD_2 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$

$$\frac{DTD_1}{DTD_2} = \frac{1.21870\text{e}{-1}}{3.85387\text{e}{-3}} = 3.16228\text{e}1$$

Calculate the GTD ratio from DTD:

$$\begin{aligned}
\frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(1.21870\text{e}{-1})^2}}{\sqrt{1-(3.85387\text{e}{-3})^2}} \\
&= \frac{\sqrt{1-1.48523\text{e}{-2}}}{\sqrt{1-1.48523\text{e}{-5}}} \\
&= \frac{\sqrt{9.851477\text{e}{-1}}}{\sqrt{9.999851477\text{e}{-1}}} \\
&= \frac{9.925\,460\,5\text{e}{-1}}{9.999\,925\,738\,1\text{e}{-1}} = 9.925\,534\,2\text{e}{-1} \\
&= 1 - 7.44658\text{e}{-3}
\end{aligned}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$\begin{aligned}
r_{s1} &= \frac{2Gm_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}32\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2} \\
&= 1.48523\text{e}5\,\text{m} \\
r_{s2} &= \frac{2Gm_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2} \\
&= 1.48523\text{e}3\,\text{m}
\end{aligned}$$

Calculate k values from the r_s/r ratio:

$$k_1 = \frac{r_{s1}}{r_1} \qquad = \frac{1.48523\text{e}5\,\text{m}}{1.00000\text{e}7\,\text{m}} \qquad = 1.48523\text{e}{-2}$$

$$k_2 = \frac{r_{s2}}{r_2} \qquad = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} \qquad = 1.48523\text{e}{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
GTD_{s1} &= \sqrt{1-k_1} = \sqrt{9.851477\text{e}{-1}} = 9.925\,460\,6\text{e}{-1} = 1 - 7.45394\text{e}{-3} \\
GTD_{s2} &= \sqrt{1-k_2} = \sqrt{9.999851477\text{e}{-1}} = 9.999\,925\,738\,1\text{e}{-1} \\
&= 1 - 7.42619\text{e}{-6} \\
\frac{GTD_{s1}}{GTD_{s2}} &= \frac{9.925\,460\,6\text{e}{-1}}{9.999\,925\,738\,1\text{e}{-1}} = 9.925\,534\,3\text{e}{-1} \\
&= 1 - 7.44657\text{e}{-3}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 7.44657\text{e}{-3} \quad (\text{via } r_s)$$

$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 7.44658\text{e}{-3} \quad (\text{via } D_{\text{crit}})$$

4.7 Case 7: (General, unconstrained)

When no constraint is applied between the systems, both mass and radius must be specified independently for each system.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.000 00e20	1.000 00e30
r (m)	2.000 00e−7	1.000 00e7

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$
$$k_1 = \frac{1.00000\text{e}20\,\text{kg}}{(2.00000\text{e}{-7}\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 7.42617\text{e}{-1}$$
$$DTD_1 = \sqrt{7.42617\text{e}{-1}} = 8.61752\text{e}{-1}$$
$$k_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-4}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-4}} = 1.21870\text{e}{-2}$$
$$\frac{DTD_1}{DTD_2} = \frac{8.61752\text{e}{-1}}{1.21870\text{e}{-2}} = 7.07107\text{e}1$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(8.61752\text{e}{-1})^2}}{\sqrt{1-(1.21870\text{e}{-2})^2}}$$
$$= \frac{\sqrt{1-7.42617\text{e}{-1}}}{\sqrt{1-1.48523\text{e}{-4}}}$$
$$= \frac{\sqrt{2.57383\text{e}{-1}}}{\sqrt{9.99851477\text{e}{-1}}}$$
$$= \frac{5.07330\text{e}{-1}}{9.999257356\text{e}{-1}} = 5.07367\text{e}{-1}$$
$$= 1 - 4.92633\text{e}{-1}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = \frac{2Gm_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}20\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}{-7}\,\text{m}$$
$$r_{s2} = \frac{2Gm_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$

Calculate k values from the r_s/r ratio:

$$k_1 = \frac{r_{s1}}{r_1} = \frac{1.48523\text{e}{-7}\,\text{m}}{2.00000\text{e}{-7}\,\text{m}} = 7.42616\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{r_2} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-4}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{2.57384\text{e}{-1}} = 5.07330\text{e}{-1} = 1 - 4.92670\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.99851477\text{e}{-1}} = 9.999257356\text{e}{-1}$$
$$= 1 - 7.42644\text{e}{-5}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{5.07330\text{e}{-1}}{9.999257356\text{e}{-1}} = 5.07368\text{e}{-1}$$
$$= 1 - 4.92632\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 4.92632\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 4.92633\text{e}{-1} \quad (\text{via } D_{\text{crit}})$$

5 Observed Conclusions

Observational, mathematical, and theoretical conclusions that were drawn from the investigation:

5.1 Time Dilation is Always a Two-System Comparison

In every case, time dilation is a comparison of the pace of time of one system relative to the pace of time of another. The relativistic nature is obscured by the current general relativity formulae. Case one demonstrates that the GTD calculation is actually a two system comparison.

5.2 Common GTD Formulae Assert Scale Privilege

Current common understandings of relativity reinforce a notion of scale privilege by imparting judgement calls about scale with operations, comparisons with abstract quantities such as flat space-time, or negligible factors. These are assertions of (human) scale privilege, because those same comparisons change meaning from the perspective of flatter space-time or a more negligible value. Flat space-time is an ideal that is approached asymptotically, but never reached. Negligibility is a matter of perspective, and is not absolute. The DTD system removes scale privilege by providing a single, unified formula for time dilation that is applicable to all systems, regardless of system geometry. This challenge is immediately alleviated by eliminating personal judgment calls, representing numbers in scientific notation form, with faithful precision, and avoiding the mindset of negligibility. There is no mathematically valid way to naturally achieve division by zero error with floating point mathematics; in computer science a sufficiently small fraction will eventually generate an out of bounds error, because it requires too many characters to represent the number. Not because you've reached zero.

5.3 Relative Inertial Density Dictates Time Dilation Between Systems

The inertial density determines the pace of time for a system about an axis or along a linear vector. Inertial density combines mass density and spatial scale into a single quantity. This investigation reveals that relative time dilation is governed by both the relative densities and the relative spatial scales of the systems. The relative pace of time is determined by both mass and density of a system, both of which are captured in DeGerlia Inertial Density as described by the equation below.

$$\text{DTD} = \sqrt{k}, \quad \text{where } k = \frac{m_1 r_2}{m_2 r_1} \tag{16}$$

$$\text{GTD} = \frac{\sqrt{1-k_1}}{\sqrt{1-k_2}}, \quad \text{where } k_1 = \frac{m_1/r_1}{D_{\text{crit}}},\, k_2 = \frac{m_2/r_2}{D_{\text{crit}}} \tag{17}$$

5.4 No Deviation from General Relativity

The principles explained in this document come directly from general relativity. The relationship between time dilation and relative spatial mass distribution of the respective systems is confirmed theoretically and mathematically to emerge from the relative inertial density of two systems about a particular axis. Given that no deviations in our observational effects are predicted, and that this equivalence emerges solely from general relativity, expected observations would always be consistent with general relativity.

5.5 No Deviations from Observational Predictions

The factors explored herein, in, and of, themselves, do not in any way change or violate how systems are observed relative to other systems. Not a single current, future, or past observation would be predicted to be different.

5.6 No Material System Exists Without Spatial Distribution

For relativistic physics, including the approximation of relativistic physics we call classical physics, every system requires spatial distribution of matter (mass). In a literal interpretation of classical physics and general relativity, a material singularity is unattainable, because such a system would lack any spatial distribution. Per the fallacy of scale privilege described in this paper, one cannot reasonably declare negligibility based on their individual perception. Further, it would be an extreme fallacy to extend this logic to reinterpret a negligible value to zero, and even more so, to leverage that chain of faulty logic to arrive at a division by zero. It must also be a mathematically valid operation. In floating-point mathematics, an infinitely precise zero is never attained except by assignment. An incredibly small fraction will eventually generate an out of bounds error, meaning there is not enough space in the data type to represent the full value. There is no fraction so small that there is not another smaller fraction; this is a notion of scale privilege.

6 Explanatory Hypotheses

For the observations where we believe we can offer explanatory insights that do not yet meet the muster of a conclusion, we have stated our hypothesis of explanation. Below we provide our explanatory hypotheses for the noteworthy observations and conclusions that can be investigated and validated through future analysis or research.

6.1 System Time Emerges from System Inertia

System time is inversely proportional to inertia about the selected axis, suggesting the time emerges from and is governed by inertia about the axis of rotation, or along the vector of motion being compared[2].

$$t \propto \frac{1}{I} \tag{18}$$

This can be validated mathematically by showing that the temporal response (delta in motion) for any system, about any axis or along any path of motion, is solely and inversely proportionally related to the inertia or moment of inertia of the system about the compared axis. Meaning that the temporal effect of torque applied or force applied is always the same delta of motion if the same moment of inertia were expressed about those dimensions. Regardless of what spatial geometry arrived at that moment of inertia about that axis.

6.1.1 The Definition of Inertia is Indistinguishable from the Definition of Time

Inertia is already defined as the sole determinant of the change in motion resulting from the application of a force to a system, regardless of the mass geometry of that system. This understanding would not change any expected observations, but rather, would offer a more complete explanation for the nature of time.

6.1.2 Like Inertia, Time is both Linear and Rotational

Inertia is both rotational (moment of inertia mr^2) and linear (mass). At any given moment, every system is moving in a single linear direction and rotating about a single axis. As such, if time emerged from inertia, it would also have two modes, both linear and rotational.

6.1.3 Like Inertia, Time is Three Dimensional

For both linear and rotational contexts, time operates in three perpendicular spatial dimensions, about an axis for rotational inertia, and along a vector for linear inertia. If time is emergent of inertia, time would also be three dimensional.

6.1.4 Causality is an Inertial Property

Causality emerges from the cumulatively conservative nature of inertia. If one subdivides an inertial system, regardless of the manner of division, as long as it's calculated along the same axis, one will always arrive at the same cumulative inertia for the system about that axis. That means the sum of the subsystems is always equal to the total of the system. This is the additivity of moment of inertia[3], where each constituent moment I_i is taken about the same axis.

$$I = \sum_i I_i \tag{19}$$

This means that the behavior of the whole is always dictated by the behavior of the constituents.

6.1.5 SR and GR Emerge Naturally and Necessarily

Perhaps the most compelling of these hypotheses is that general and special relativity emerge naturally and necessarily if one presumes that time behaves as described by this hypothetical explanation. General relativity corresponds to a moment of inertia (rotational inertia), whereas special relativity corresponds to linear inertia. SR is linear time dilation. GR is rotational time dilation.

6.2 Formulaic Similarities Further Link SR and GR

The shared quantitative form of general relativity versus special relativity suggests a unified structure. Both GR and SR share a unified form: $\sqrt{1-k}$, where k is a ratio of characteristic lengths or a "linear scale factor". In the case of special relativity, $k = v^2/c^2$ is a ratio of characteristic lengths traveled per second, for example, per static unit of time. k is a linear scale factor across all forms of time dilation described herein.

6.3 DTD Accurately Predicts Asymmetrical Systems

DTD effectively serves as a unified model between Schwarzschild and asymmetrical black hole systems, such as Kerr black holes.

This hypothesis can be validated by calculating a system about three perpendicular axes, treating each dimension as an individual Schwarzschild solution about that axis using D (m/r) and comparing with the Kerr field equation.

6.3.1 Frame Dragging Stems from an Incorrect Presumption

The notion of frame-dragging stems from the presumption that any system with two different radii must therefore be spinning about a single axis. This logic has then been extended to arrive at a presumption that results in this hypothesized time dilation paradox. From a static observation, one could never discern whether there are one, two, or three unique radii in the system. But, without speculating about whether these systems may or may not be spinning about an axis at some appreciable pace, we can safely say that every system in the universe, "spherical" or otherwise, is asymmetrical to some degree. A galaxy or even a cosmic filament; those systems are far from symmetrical and uniform. Yet they have properties that can be accurately modeled with this framework. They just have unique properties about every axis.

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity_STE_cases`.

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